

Mentor Scoring System – Ideation Document

Description

The objective of this project is to evaluate and rank mentors on different aspects such as their mentees' progress, engagement, and feedback received in a fair and structured manner.

Mentor Score (M)

The overall mentor score is calculated by

$$M = w1 * P + w2 * R + w3 * E + w4 * F$$

where:

- P = Student Progress Score
- R = Responsiveness Score
- E = Engagement Score
- F = Feedback Score

and the respective weights are:

- $w1 = 0.27$
 - $w2 = 0.25$
 - $w3 = 0.32$
 - $w4 = 0.16$
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Core Components

(a) Student Progress Score (P)

It is defined as the ratio of milestones completed to total milestones assigned to all the mentees of a particular mentor.

$$P = \text{completed} / \text{total}$$

This is the result of both mentees' efforts and mentor's guidance.

(b) Responsiveness Score (R)

It gives the mentor a score based on their response time to student's queries. Higher the average time, lower the score and vice versa.

$$R = e^{-(t_{\text{avg}} / 4)}$$

An exponential equation is used because a higher time like 8-10 hours really deserves a very low score (~0.1) and a faster response of like 1-2 hours deserves a quite high score (~0.65).

(c) Engagement Score (E)

It measures mentor involvement by following parameters:

- Meetings conducted
- Code reviews
- Messages

Each component was normalised per mentee and then scaled

- Meetings normalised by 5
- Code reviews by 5
- Messages by 15

Final Engagement Score is calculated as

$$E = 0.35 * \text{meet} + 0.35 * \text{code} + 0.3 * \text{msg}$$

Meetings and code reviews are given higher importance because they involve direct interactions and technical guidance with mentors while messages are often less detailed.

(d) Feedback Score (F)

Feedback is given by mentees on the scale [1-5] which is normalised to [0-1]

To reduce any kind of extreme ratings, the reviews are clamped between 1.5-4.5

This would minimise the highly positive or highly negative ratings.

Weight Justification

Engagement is assigned the highest weight as it directly involves the mentor's activeness towards their mentees through meetings and reviews. Progress is given a decent importance because it shows the result of his mentorship. It has less importance than engagement because progress not only requires mentor's guidance but mentees' efforts too. Responsiveness has a moderate rating because it reflects in his availability for his mentees. Feedback has the lowest because it is quite similar to progress because if a student performs well, he'll probably give a good rating and the opposite for not performing well. So this is dependent on student's progress.

Score evaluation over time

Mentor's performance is updated weekly by

$$M_{\text{new}} = 0.7 * \text{current} + 0.3 * \text{previous}$$

This will prioritize the recent activities while still considering his past activities.

Activity Decay

If a mentor is inactive for over 2 weeks, his score is reduced as

$M_{\text{new}} = M_{\text{old}} * (1-d)$ where $d = 0.1$

This ensures losing score over inactivity.

Normalisation across mentees

Every score was normalised per mentee to ensure that mentors with a large number of mentees are neither advantaged nor penalized.

Assumptions

- Feedback ratings are reliable after reducing bias.
 - Each mentee has equal contribution to their mentor's evaluation.
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Conclusion

This scoring system uses multiple aspects such as engagement, feedback ,etc. to rank mentors. It tries to ensure fairness and does not support any bias.